ADEQUACY IS PRESERVED BY RESTRICTION ALONG A RING MAP

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The presentation of an adequate functor as a kernel of a map of module-valued functors. The definition of adequate functors, invoked at proof line 960. Supplies the exact sequence used.
- Base change of modules: M tensor_A B' = (M tensor_A A') tensor_{A'} B'. Standard module base change, used at proof lines 962-963. Identifies the presenting module over A'.

Lemma 0.1. *Let $A \to A'$ be a ring map. If F is an adequate functor on Alg_A, then its restriction F' to $Alg_{A'}$ is adequate too.*

Proof. Choose an exact sequence $0 \to F \to \underline{M} \to \underline{N}$. Then $F'(B') = F(B') = \mathrm{Ker}(M \otimes_A B' \to N \otimes_A B')$. Since $M \otimes_A B' = M \otimes_A A' \otimes_{A'} B'$ and similarly for N we see that F' is the kernel of $\underline{M \otimes_A A'} \to \underline{N \otimes_A A'}$. □

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